Introduction to Machine Learning. Machine learning is a field of study in which computers improve their performance on a task from experience, without being explicitly programmed for every case. The three main paradigms are supervised learning, unsupervised learning, and reinforcement learning. In supervised learning, each training example pairs an input with a known label, and the model learns a mapping from inputs to labels. Classification predicts a discrete category, such as whether an email is spam, while regression predicts a continuous value, such as a house price. In unsupervised learning, the data has no labels, and the algorithm looks for structure, for example by grouping customers into clusters. Clustering and dimensionality reduction are the most common unsupervised tasks. In reinforcement learning, an agent takes actions in an environment and receives rewards, and it learns a policy that maximizes the total reward over time.

Generalization and the bias-variance trade-off. A model is useful only if it performs well on data it has not seen during training. The training error measures performance on the data used to fit the model, while the test error measures performance on held-out data. A model with high bias makes strong assumptions and underfits, meaning it fails to capture the true pattern even in the training data. A model with high variance is very sensitive to the particular training sample and overfits, meaning it memorizes noise. Overfitting is usually diagnosed when training error keeps falling while validation error starts to rise. Common remedies include collecting more data, simplifying the model, adding regularization, and stopping training early.
K-fold cross-validation splits the data into K parts, trains on K minus one of them, and evaluates on the remaining part, repeating this K times so that every example is used for validation exactly once.